2. **Scatter Plot:** A visual representation of the relationship between height and weight. This plot enables you to identify trends, such as whether taller individuals tend to weigh more, and spot outliers that deviate significantly from the trend.

Extending the Script

To further enhance this analysis, consider the following extensions:

1. **Data Cleaning:**
    - Check for missing values and handle them using techniques like mean imputation or removal of incomplete rows.

```
if data.isnull().values.any():
    print("Missing values found. Filling with mean...") data.fillna(data.mean(),
inplace=True)
```

2. **Adding New Columns:**
    - Create additional columns for derived metrics, such as Body Mass Index (BMI).

```
data['BMI'] = data['Weight'] (data['Height'] 39.37)**2
print(data.head())
```

3. **Advanced Visualizations:**
    - Use Matplotlib or Seaborn to create histograms, box plots, or regression plots to explore data further.

```
import seaborn as sns
sns.regplot(x="Height", y="Weight", data=data, scatter_kws={"color": "blue"},
line_kws={"color": "red"}) plt.title("Height vs Weight with Regression Line") plt.show()
```

4. **Saving the Output:**
    - Save the cleaned dataset and generated plots for future use.

```
data.to_csv("cleaned_data.csv", index=False) plt.savefig("height_vs_weight_plot.png")
```

This script demonstrates the fundamental steps in data analysis, from loading a dataset to visualizing relationships between variables. By incorporating additional features such as data cleaning, derived metrics, and advanced visualizations, you can perform a comprehensive analysis that lays the groundwork for more advanced ML tasks. Mastery of these tools and techniques will ensure a strong foundation in Python-based data science.